

Co-4 AI-7

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* Comparative analysis

1) * Compare Hashing and binary Search based on time Complexity and methods

2) Compare the time Complexity of all notations. in Bubble, insertion, Selection, merge, Quick, Heap.

1) Comparing between Hashing and binary Search

Basis	Hashing	Binary search
Definition	Hashing stores data using a hash function that maps keys to position in a hashtable	Binary search finds an element by repeatedly dividing a sorted data set into halves
Searching method	Uses a hash function to directly calculate the probable location of an element.	Compares the target with the middle element and searches either the left or right half.
Data required	Data does not to be sorted	Data need to be sorted.
Best Case Complexity	$O(1)$	$O(1)$

Avg Case
Complexity

$O(1)$

$O(\log n)$

Worst
Case
Complexity

$O(n)$ due to
Collision

$O(\log n)$

Insertion

$O(1)$ average

$O(n)$ for an
array.

Deletion

$O(1)$

$O(n)$

space

$O(n)$

$O(1)$

main
issue

Hash Collision
may occur

Requires
sorted data.

Common
applications

Dictionaries,
Symbol tables,
Databases.

Searching
Sorted arrays
and lists.

Property	Bubble sort	Insertion sort	Selection sort	merge sort	Quick sort	Heap sort
Best Case	$O(n)$	$O(n)$	$O(n^2)$	$O(n \log n)$	$O(n \log n)$	$O(n \log n)$
Avg Case	$O(n^2)$	$O(n^2)$	$O(n^2)$	$O(n \log n)$	$O(n \log n)$	$O(n \log n)$
Worst Case	$O(n^2)$	$O(n^2)$	$O(n^2)$	$O(n \log n)$	$O(n^2)$	$O(n \log n)$
Space Complexity	$O(1)$	$O(1)$	$O(1)$	$O(n)$	$O(\log n)$	$O(1)$
Stable	yes	yes	No	yes	No	No
In-place Method	yes	yes	yes	No	yes	yes
Main drawback	Comparison (Comparison)	Comparison (Comparison)	Comparison (Comparison)	Divides & Conquer	Divide & Conquer	Heap based
	Simple	Efficient for nearly sorted	Reverses	Guaranteed	Fast average case	Guaranteed
				$O(n \log n)$		$O(n \log n)$

